Hawaiʻi's Energy Future

Ethan Hartley   Michael J. Roberts

University of Hawaiʻi at Mānoa — Economics, Sea Grant, UHERO

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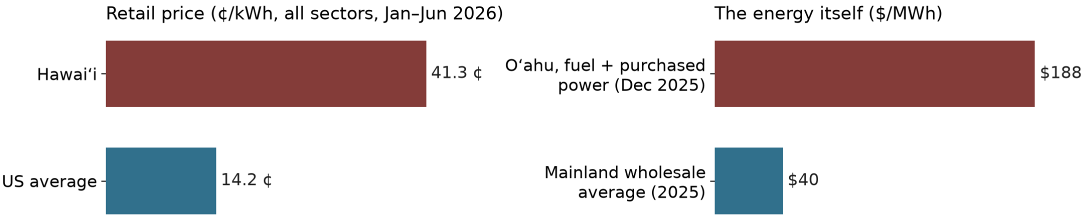
All code, data, and results: `github.com/mikejrob/oahu-electricity-future`

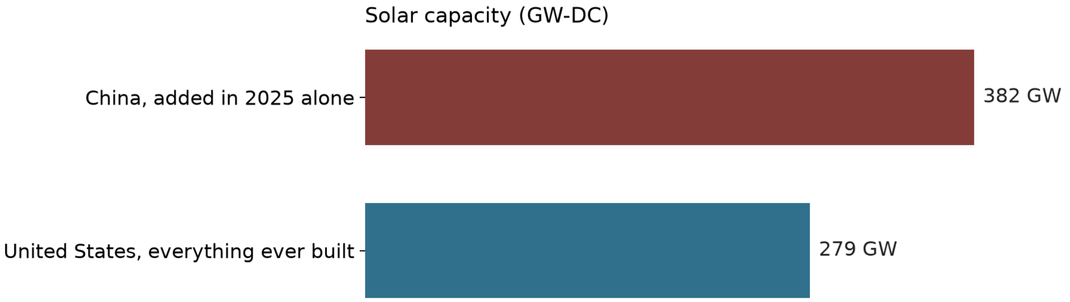
Highest prices in the nation — and the first 100% mandate



EIA, Electric Power Monthly; Hawaiian Electric, O'ahu energy-cost filings (Aug 2026: $258 after this year's oil spike); EIA, major-hub day-ahead average.

- First state to mandate 100% renewable electricity (2015, by 2045)
- Ten years in, the grid is still ∼2/3 oil: about 33–37% clean vs. a 43% U.S. average

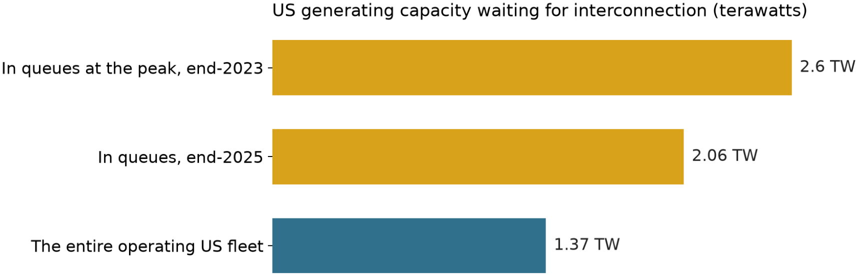
Clean energy is taking over — globally



SolarPower Europe, Global Market Outlook 2026; SEIA/Wood Mackenzie, Year in Review 2025.

- Spot LNG roughly doubled since the Hormuz closure (\$11–13 $\rightarrow$ \$22–23/MMBtu); diesel cracks at all-time records
- Electric vehicles: $\sim$2/3 of recent new-car sales in China, $\sim$2/3 of India's three-wheeler sales

The U.S. bottleneck: connecting to the grid



LBNL, Queued Up, 2024 and 2026 editions. Solar, wind, and batteries: 95% of the end-2023 queue, 85% of end-2025.

- Median wait: 5+ years for projects finished in 2025, under 2 in 2008; of capacity queued 2000–2020, 13% was operating by end-2025
- Panels are ∼25% of a U.S. utility solar project (∼16% with storage); field labor ∼22% and design, overhead & other soft costs ∼16% of the total (NREL 2025)
- IRENA 2025: U.S. installed cost 2.2× China's ($1,180 vs. $533/kW); China is cheaper on 13 of 16 components, and the panel gap is one of the smaller ones

Hawaiʻi has the problem in its sharpest form

- **Rooftop solar**: the nation's largest share, $\sim$18% of sales (California next, $\sim$13%) — yet daytime exports earn 10–13.5 ¢ by tranche, roughly half the filed avoided energy cost ($\sim$24 ¢) and under a third of retail

- **Utility solar**: highest prices in the country ($\sim 2\times$ mainland) — and interconnection as slow as the mainland's, without the queue

- **The response on offer**: new oil and LNG plants

The crux: make it easy to connect to the grid and sell power at a fair price.

Three decisions are on the table now

- **Waiau Repower —** six oil-fired turbines, approved March 2026
- **JERA proposal —** $2B, 500 MW plant + LNG import terminal
- **Utility-scale solar —** procurement has stalled; land rules *may* bind

Each choice locks in decades.

What we built

- Open capacity-expansion model of O‘ahu, 2027–2050 — built on Switch (Fripp), updated and extended
- Hourly operations: unit commitment, reserves, storage, EVs
- 500+ solved scenarios: oil prices, solar costs, land rules, LNG, geothermal
- Everything public — inputs, code, results, an interactive explorer; anyone can replicate it or test their own scenarios

How the model works

The model in words

– **Decides**: what to build and when — solar, wind, batteries, geothermal, plants — and how to run everything, hour by hour

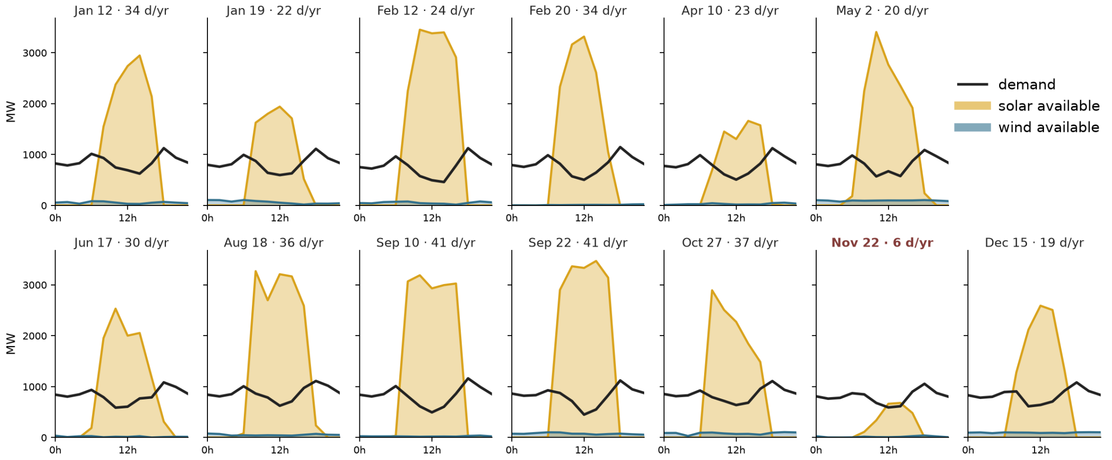
– **Objective**: lowest total cost of electricity through 2050 (capital, fuel, and operations)

– **Always**: demand met every hour; reserves held for the largest unit running, plus swings in wind and solar

– **Limits**: land for solar, build schedules, existing plants and contracts, 100% renewable by 2045

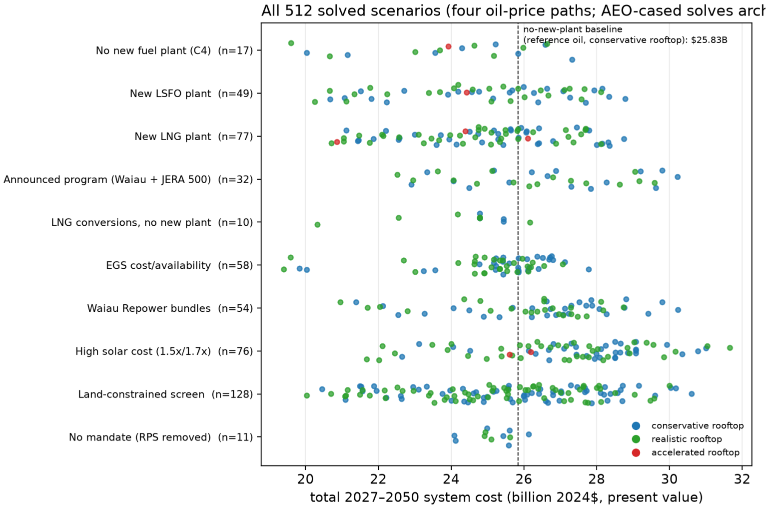
– **Flexibility**: batteries, 50% of EV charging, and hydrogen production scheduled to the cheapest hours

Each model year: 13 sample days, weighted



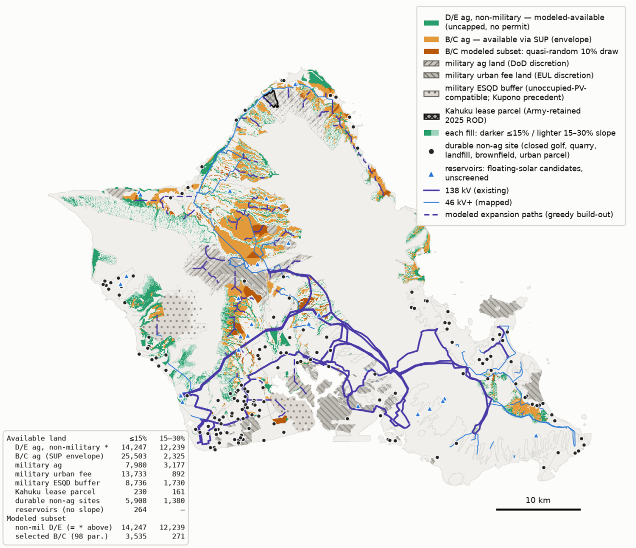
2045, no-new-plant base case. Solar and wind at full availability from the capacity the model built; demand is net of rooftop solar and excludes flexible EV and hydrogen loads, which the model schedules. Nov 22: the record's hardest day.

The scenario space



Solar and land

Plausibly available land, and what the model uses



Available land	≤15%	15–30%
D/E ag, non-military *	14,247	12,239
B/C ag (SUP envelope)	25,503	2,325
military ag	7,980	3,177
military urban fee	13,733	892
military ESQD buffer	8,736	1,730
Kahuku lease parcel	230	161
durable non-ag sites	5,908	1,380
reservoirs (no slope)	264	–
Modeled subset		
non-mil D/E (= * above)	14,247	12,239
selected B/C (98 par.)	3,535	271

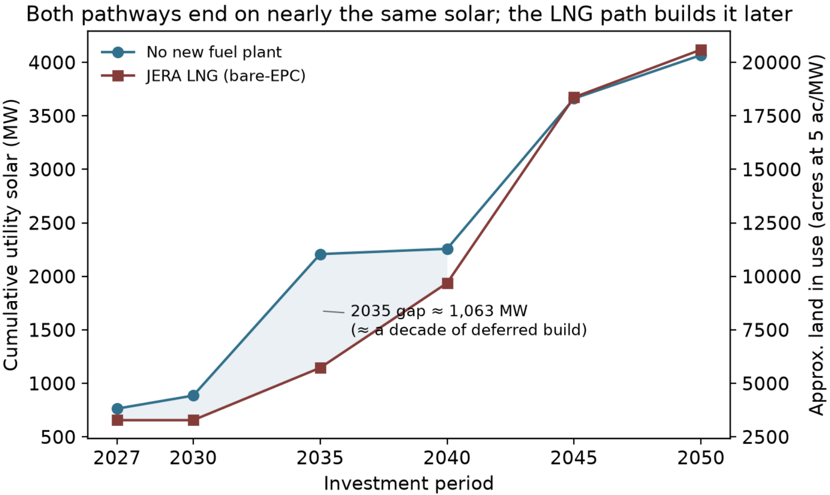
Each fill has two shades: darker $\leq$15% slope, lighter the 15–30% increment.

The modeled subset is non-military D/E agricultural land (green) plus a quasi-random 10% draw of B/C (dark orange).

Military land is shown by tenure — available at DoD discretion, not in the modeled subset.

No grid-distance filter; published near-grid figures are smaller.

How much land, and when



Rooftops change the land question

- Measured potential: about 4,000 MW on existing structures
- Three adoption paths: $\sim$1,000 / 1,600 / 2,100 MW by 2050
- Fast rooftop growth: utility solar falls from 4,100 to 3,000 MW; land toward 15,000 acres
- What unlocks it: paying fair value for exports (Section 2.7)

Hawai‘i pays a large, unexplained solar premium

Utility solar here: 1.4 to 2.2+$\times$ mainland cost

- Physical conditions explain only part
- 2018-era contracts (Phase 1, Kaua‘i) came in *below* 1.8$\times$
- Recent O‘ahu repricings anchor the top of the range

The premium is in the process, not the hardware

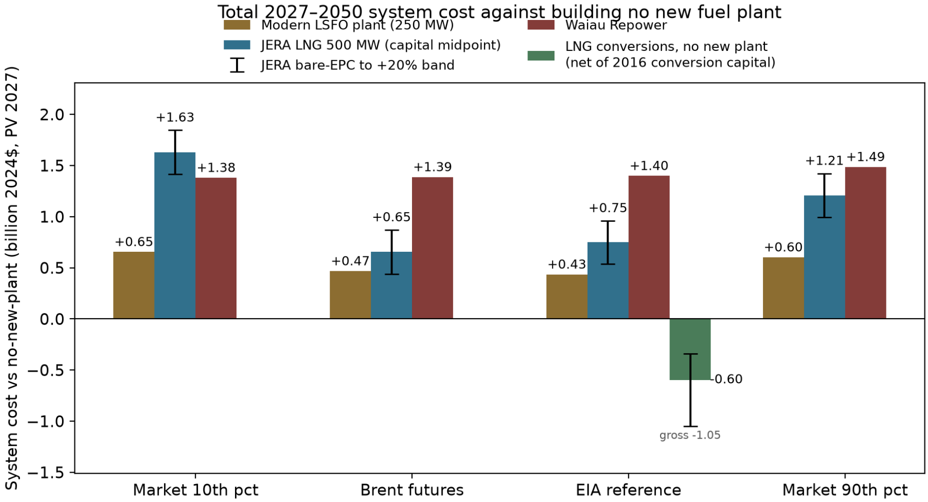
Utility solar + storage	Hawaiʻi	Mainland
Contracts of 2018–20	7.7 ¢	3.1 ¢
Contracts of 2024–26	19.5 ¢	6.5 ¢
The same global shock added	+12 ¢	+2.5–3.5 ¢

Levelized 2024¢/kWh, matched configuration. The 12¢ is Mahi — the *same project*, repriced. Meanwhile grid-battery packs fell **45% in 2025 alone**, to $70/kWh (BloombergNEF).

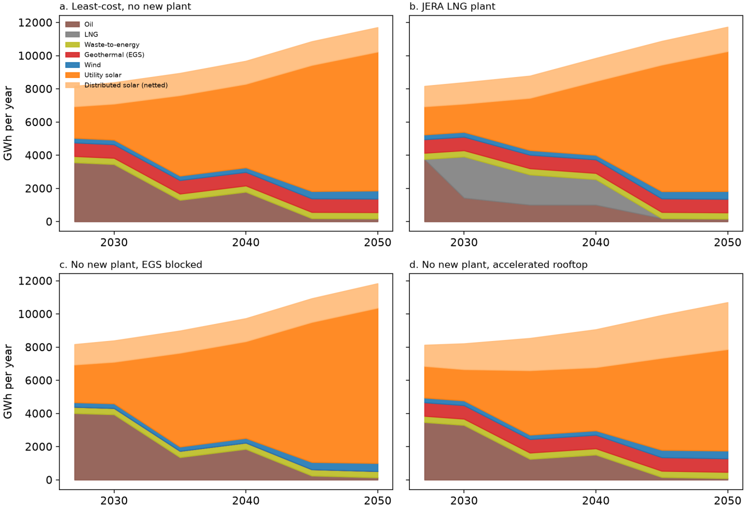
- **Rooftop**: modest. Tesla's by-state pricing (8 kW+): Hawaiʻi $2.90/W, between California ($2.83) and New York ($2.98)
- **Utility-scale**: 2–3×. Similar panels, labor, and land — most of the difference appears to be soft costs
- **One mechanism**: a PPA fixes its price in nominal dollars at award, and award-to-operation runs **4–7 years** on Oʻahu. Inflation eats the award, so developers bid high, reprice, or walk

Does a new fuel plant pay?

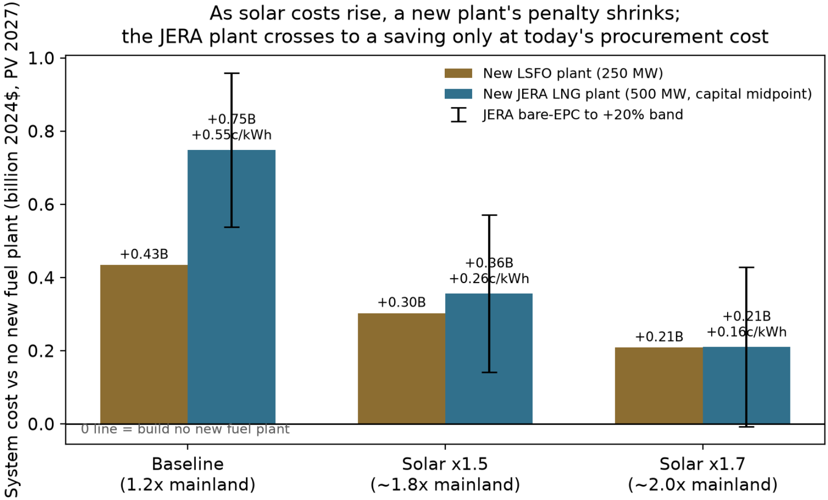
No proposed plant beats building no plant



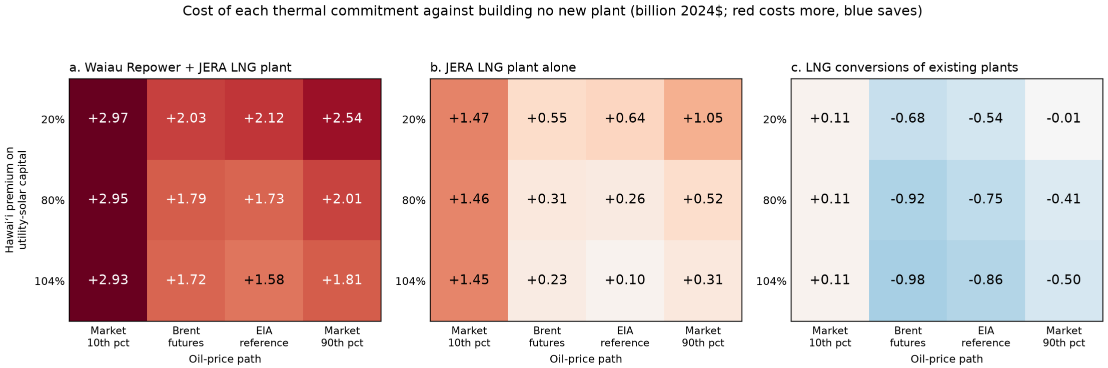
What the system builds instead



The verdict survives expensive solar



…and every oil price



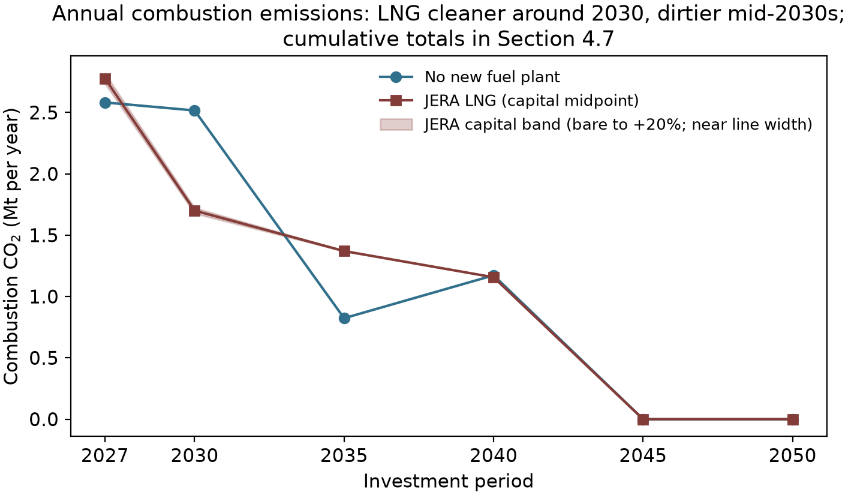
If LNG comes, existing plants beat new ones

LNG configuration (vs. no new plant)	Δ cost
New JERA plant (midpoint)	$+\$0.7$–$1.6$B
Convert Kalaeloa alone	$-\$0.4$B
+ Kahe 5 & 6, CIP turbine	$-\$1.1$B

Same fuel savings, no construction. Conversion budgets: ~\$1,800/kW before the case breaks.

And the LNG itself is a bet: a long-term *quantity* contract, likely with a price floor. We model \$11.4/MMBtu delivered — today's spot is about twice that.

LNG’s climate case hangs on methane leakage



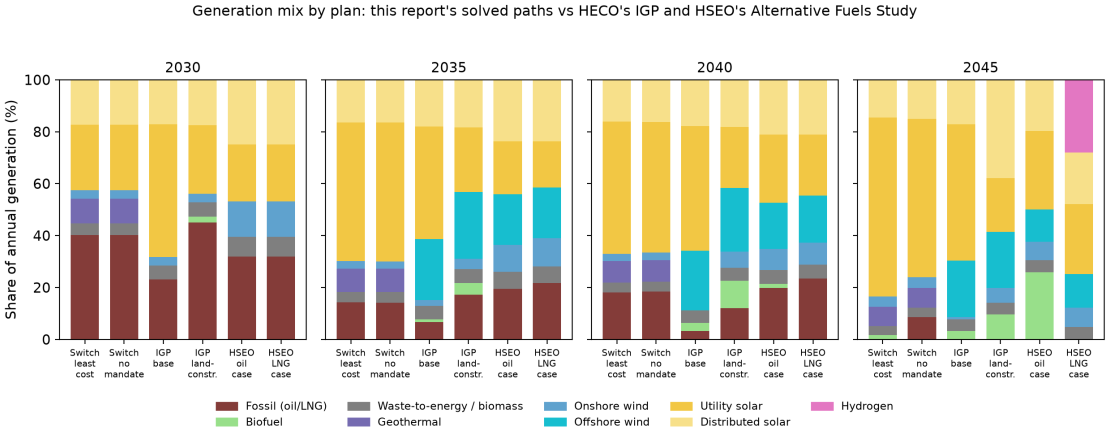
The leakage arithmetic

Greenhouse advantage gone above $\sim$0.9% supply-chain leakage

- (100-year basis; $\sim$0.3% on a 20-year basis)
- Measured U.S. supply-chain rates sit well above both thresholds

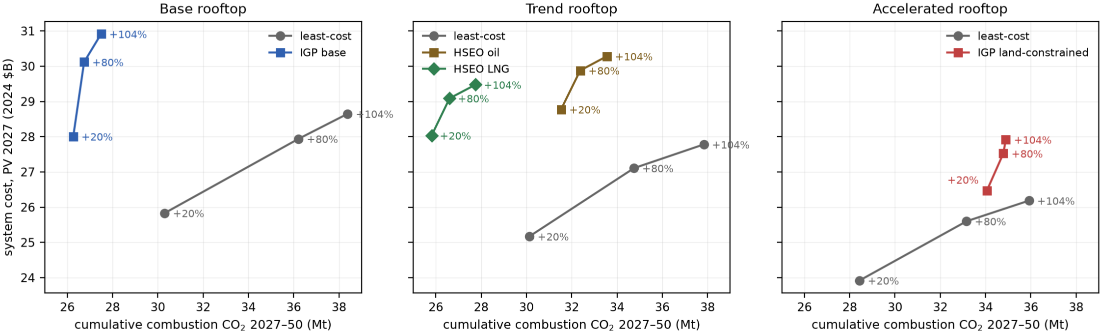
The plans on file

Our least-cost paths vs. HECO's IGP and HSEO's study



Every plan, priced on one ruler

Published plans vs the least-cost path as the solar premium rises: plans hold their mix (near-vertical); the optimum trades carbon for cost



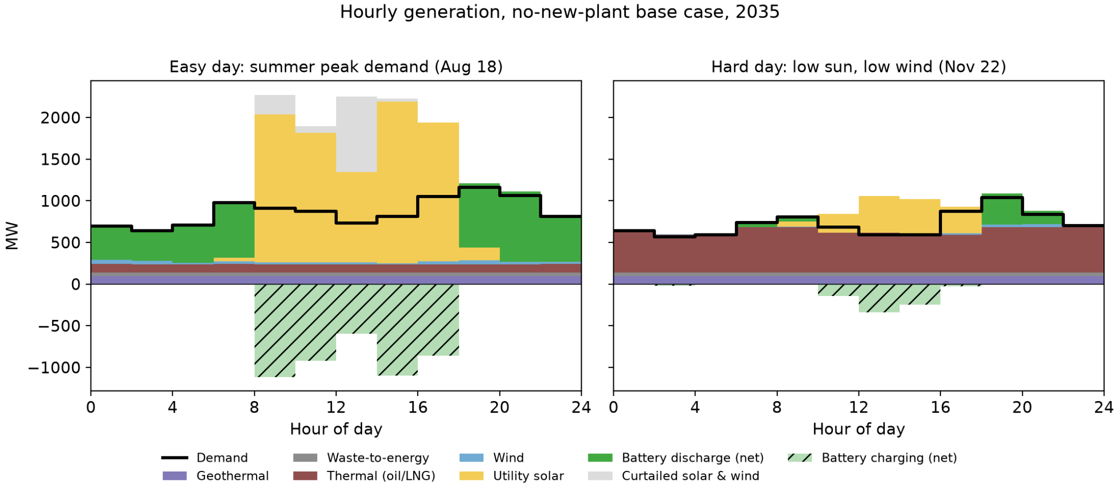
Enhanced geothermal

A small resource with outsized value

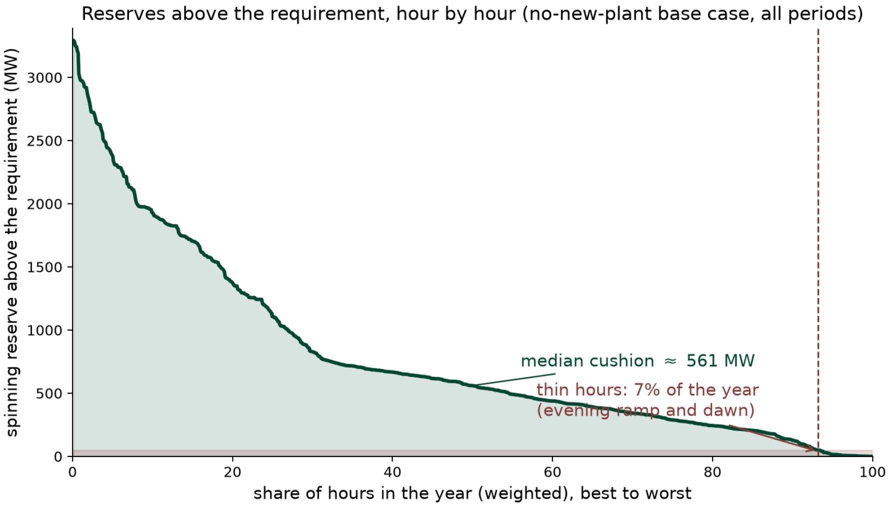
- First resource screen: $\sim$100 MW identified on O‘ahu
- The model builds all of it in the base case
- Blocking it costs +\$0.6B; at optimistic drilling costs the saving reaches $\sim$\$1.0B
- 100 MW displaces $\sim$400 MW of solar land
- The resource at depth is unmeasured — could be far larger

Reliability

The hard day is not the peak day



Resource adequacy: covered with room to spare



Resilience changes on a renewable grid

- Firm plants cover the daily-average net load ($\sim$610 MW in 2035); batteries carry energy to the evening peak ($\sim$1,270 MW)
- Generation comes in thousands of pieces — the main risk becomes island-wide, multi-day, forecastable weather
- Two units out at a thin hour, at the filed 10% failure rate: $\sim$6 h/yr — conventional standards run 2–8 h/yr
- The real tail: correlated failures and storage depth (a 16-day low-wind run is on record) — v2 tests both

The Waiau Repower fails its own cost test

- Adds +$1.4B to system cost on every oil path
- Runs at 51% in 2030, under 32% by the late 2030s, $\sim$0 from 2045
- A modern combined cycle delivers the same firm capacity at $\sim$1/3 less per kW
- PUC cost cap: $220–310M of the bill lands on shareholders

What reform unlocks

Three levers, no new technology required

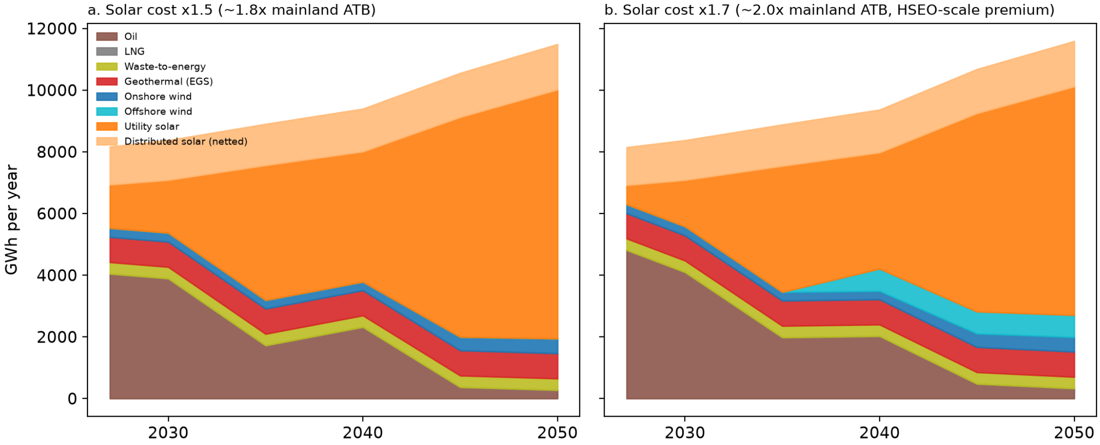
- **Land**: state siting rules exclude most of the island's flat, cleared, transmission-adjacent land
- **Procurement**: independent bid evaluation, standard contracts, published queues
- **Rooftops**: pay fair value for exported power

The savings at stake dwarf every supply-side proposal on the table.

The deeper reform: retail wheeling

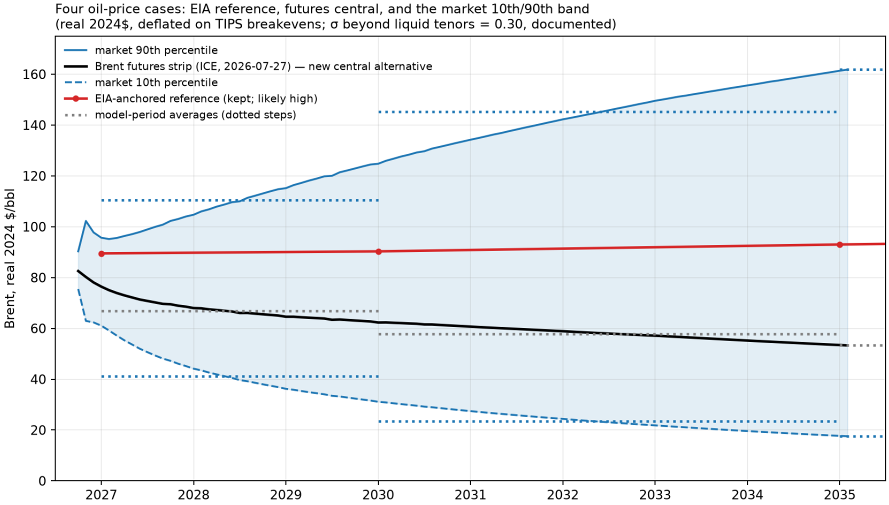
- The Legislature has mandated retail wheeling — *how* is the open question
- Our proposal (Schedule RWS, in preparation): sellers and large buyers settle at **locational marginal prices**; the utility keeps the wires
- Ten-day provisional interconnection replaces the RFP gauntlet
- Real-time prices unlock demand flexibility (Imelda, Fripp & Roberts 2024, *AEJ: Policy*)
- Self-funding: a transition fund captures part of today's oil-priced margins

Even if solar stays expensive here, the answer holds

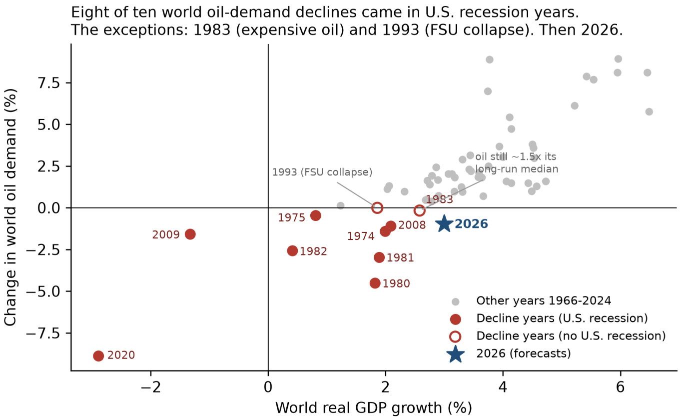


The oil bet

Four oil futures, priced from markets



Demand is falling in 2026 — without the recession that always came before



Sources: EI Statistical Review 2025 (oil, consumption basis, 1966-2024); World Bank WDI (GDP, market exchange rates); U.S. recession years from NBER dating (FRED USREC) - world recessions have no comparable dating. 2026: IEA OMR July 2026 (-1.0 mb/d) and IMF WEO Update July 2026 (3.0% growth, PPP weights; ~2.4% market-rate-comparable).

Takeaways

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- **No proposed fuel plant pays for itself** — not JERA's, not the Waiau Repower, under any oil or solar price we can defend
- **Solar + storage + existing plants** is the robust least-cost core; geothermal sweetens it
- **The binding constraints are rules, not resources** — land, procurement, and rooftop compensation

Kick the tires

`github.com/mikejrob/oahu-electricity-future`

- Full report, all inputs, all 500+ solved scenarios
- Interactive explorer — every figure, every configuration
- Comments to September 15, 2026; we answer publicly

Backup

System cost by trajectory and oil price (Table ES.1)

	Market 10th	Brent futures	EIA ref.	Market 90th
No new fuel plant	21.2	24.3	25.8	27.3
Modern LSFO plant	+0.7	+0.5	+0.4	+0.6
JERA LNG (midpoint)	+1.6	+0.7	+0.8	+1.2
Waiau Repower	+1.4	+1.4	+1.4	+1.5
Waiau + JERA	+3.1	+2.1	+2.2	+2.7

Billions of 2024$, present value 2027–2050, 3% real.

Method guardrails

- 0.1% optimization tolerance; dominance checks across 698 scenario pairs gate every release
- 13 sample days per period include the record's hardest day (Nov. 22, 2008: low sun, weak trades)
- Reserves every hour: the largest unit online, plus regulation that grows with solar and wind output
- Conservative for renewables: no inter-day storage, no real-time pricing, distributed value uncredited

Since the draft: Kalaeloa signed for 30 more years

- New PPA signed July 27, 2026 (pending PUC approval)
- 208 MW firm, biofuel-capable repowering, $\sim$2033–2063
- Capacity charge *fell* $7/kW-yr — firm capacity without new construction keeps beating new plants